

# Hydrological Studies of Atlur Mandal in YSR District, Andhra Pradesh using SRTM Data and GIS Techniques

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This study attempts to study the hydrological characteristics of Atlur area sub-watershed using geoprocessing techniques in GIS. This technique is found relevant for the extraction of study area and its drainage networks. The extracted drainage network was classified according to Strahler's system of classification and it reveals that the terrain exhibits dendritic to sub-dendritic drainage patterns. The study area was of fifth order to lower order streams with a drainage density value of 1.48 km/km<sup>2</sup>. The slope of basin varied from 0-45° and the slope variation is chiefly controlled by the local geology and erosion cycles. The elongation ratio of the basin is 0.63 indicating that the study area is elongated with moderate relief and steep slopes. The drainage texture of the basin is 3.1 indicating an intermediate texture that exists over the region. Remote sensing data (SRTM-DEM) coupled with geoprocessing techniques prove to be a competent tool used in morphometric analysis and evaluation of linear, slope, areal and relief aspects of morphometric parameters. Moreover, the data can be used for basin management and other hydrological studies in future.

## KEYWORDS

Hydrological studies, Morphometric parameters, Atlur mandal, SRTM data, ArcGIS

## 1. INTRODUCTION

Morphometry is defined as the measurement and mathematical analysis of the configuration of the Earth's surface and the shape and dimensions of its landforms. Morphometric analysis of river basins provides a quantitative description of the drainage system, which is an important aspect of the characterization of basins [1]. It is important in any hydrological investigation, like assessment of groundwater potential, groundwater management, basin management and environmental assessment. The development of a drainage system over space and time is influenced by several variables, such as geology, structural components, geomorphology, soil and vegetation of an area through which it flows. Various hydrological phenomena are correlated with the physiographic characteristics of a drainage basin, such as size, shape, slope of the drainage area, drainage density, size and length of the tributaries, etc., [2,3]. The dynamic nature of runoff is controlled by the geomorphologic structure of the catchment area and the induced runoff is very sensitive to the morphometric characteristics of the contributing area [4]. The morphometric analysis is per-

formed through measurement of linear, aerial, relief, gradient of channel network and contributing ground slope of the basin [3,5,6]. In India, National Institute of Hydrology (1993) carried out hydro-geomorphological studies of various basins and their analyses are based on linear, aerial and relief aspects using different mathematical equations. Various morphometric parameters, such as drainage pattern, stream order, bifurcation ratio, drainage density and other linear aspects are studied using remote sensing technique and topographical map [7]. The surface runoff and flow intensity of the drainage system are estimated using the geomorphic features associated with morphometric parameters [8]. Pioneer work on basin morphometry has been carried out by Horton, Miller, Smith and Strahler [1,9-12].

Application of remote sensing provides a reliable source for the preparation of various thematic layers for morphometric analysis. The digital elevation data is used for generating the elevation model of a landscape to any extent. The resolution of the image may vary with respect to the satellite sensors. The processed DEM is used for generating the stream network and other supporting layers [3,13]. Geographical information systems (GIS) have been used for assessing various basin parameters, providing flexible environment and powerful tool for determination, interpretation and analy-

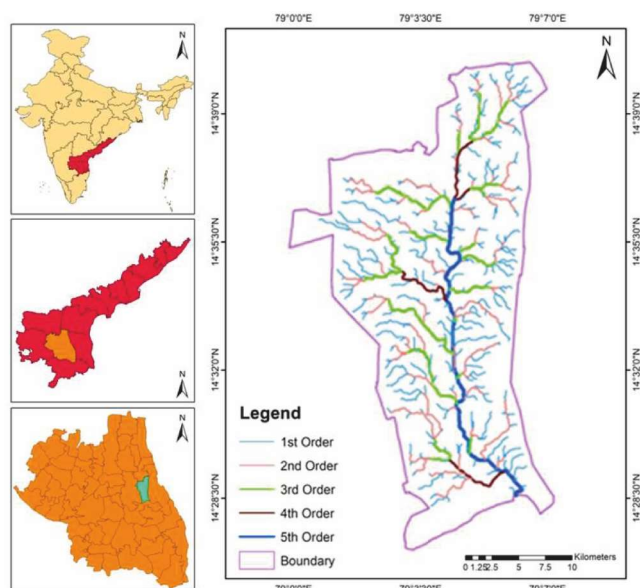


Figure 1. Location map of the study area

sis of spatial information related to river basins. Geology, relief and climate are the primary determinants of a running water ecosystem functioning at the basin scale [14]. Thus, detailed study of morphometric analysis of a basin is of great help in understanding the influence of drainage morphometry on landforms and their characteristics. The present study area emphasizes various morphometric parameters of Atlur area subwatershed in YSR district, Andhra Pradesh using remote sensing and GIS techniques.

### 1.1 Study area

The study area lies between latitudes  $14^{\circ}26'57.9336''$  to  $14^{\circ}40'38.7726''$  north and longitudes  $78^{\circ}59'45.819''$  to  $79^{\circ}7'31.458''$  east and falls in the Survey of India toposheet no. 57 N/2, N/3 covering an area of  $179.96 \text{ km}^2$ . Location map of the study area is shown in figure 1. The study area is sub-watershed boundary and stream network of basin was obtained from shuttle radar topography mission (SRTM) data. The extracted basin and stream network were projected to the regional projection (WGS 1984 UTM Zone 44N). The major litho units observed are the Nallamalai group and overlying Chitravathi group consisting of Bairenkonda quartzite formation at the bottom and cumbum shales at the top. The Bairenkonda quartzites are brown, pink, grey or greenish in colour and at few localities they are ferruginous. They exhibit ripple marks and current bedding. Nallamalai group occurs to the east and north of the Archaean and Dharwar. The cumbum formations comprise shales and phyllites of variegated colours. Numerous bands of limestones and quartzites occur



Figure 2. Geology map of the study area

as intercalations. The study area consists of Nagari quartzites and shales with phyllites [15]. Geology map of the study area is shown in figure 2.

## 2. MATERIAL AND METHOD

The shuttle radar topography mission (SRTM) data downloaded from USGS site was used to generate the most complete high-resolution digital topographic database of Earth with a spatial resolution of 90 m. Using ArcGIS 10.4 the SRTM data was processed for extraction of the drainage network. Digitization work has been carried out for entire analysis of basin morphometry using the resulted output raster from SRTM. The order was given to each stream by following Strahler's stream ordering techniques [1].

## 3. RESULT AND DISCUSSION

### 3.1 Morphometric analysis

The present study integrated the use of remote sensing and GIS techniques in morphometric analysis and the results of morphometric parameters were discussed. The study of basin morphometry attempts to relate basin and stream network geometries to the transmission of water and sediment through basin. The

**Table 1. Stream order, number and bifurcation ratio of stream orders**

| Stream order (U) | Number of streams ( $N_u$ ) | Bifurcation ratio ( $R_b$ ) |      |
|------------------|-----------------------------|-----------------------------|------|
| I                | 285                         | 1st/2nd order               | 4.13 |
| II               | 69                          | 2nd/3rd order               | 3.63 |
| III              | 19                          | 3rd/4th order               | 4.75 |
| IV               | 4                           | 4th/5th Order               | 1    |
| V                | 1                           | —                           | —    |
| Total            | 378                         | Mean bifurcation ratio      | 3.37 |

size of a drainage basin acts upon the amount of water yield, the length, shape and relief, affecting the rate at which water is discharged from the basin. The morphometric analysis was carried out with respect to the parameters, like stream order, stream length, bifurcation ratio, stream length ratio, basin length, drainage density, stream frequency, elongation ratio, circularity ratio, form factor and relief ratio using mathematical formulae given in table 1 and results are given in table 2.

### 3.2 Linear aspects of the channel system

**3.2.1 Stream order (U) :** The first step in the drainage basin analysis is designation of stream orders which is not only the index, the size and scale, but also to afford an approximate index of the amount of stream flow, which can be produced by a particular network, stream order and number which is directly proportional to the size of the contributing watershed. It is observed that there are five orders of streams in study watershed (Table 1).

**3.2.2 Stream number ( $N_u$ ) :** The count of stream channels in its order is known as stream number. The number of stream segments decreases as the order increases. The higher amount of stream order indicates lesser permeability and infiltration. Stream number is directly proportional to size of contributing watershed and to channel dimensions. It is obvious that the number of streams of any given order will be fewer than for the next lower order but more numerous than for the next higher order. In the present study the order-wise stream numbers estimated are given in table 1.

**3.2.3 Bifurcation ratio ( $R_b$ ) :** Bifurcation ratio is defined as number of streams of one order to the next higher order. Bifurcation ratios characteristically range between 3 and 5 for watershed in which geologic structures do not distort the drainage pattern [16]. Bifurcation ratio is mainly controlled by the basin shape and

not only influences the landscape and morphometry but also controls the surface runoff. The bifurcation ratio will not be precisely the same from one order to the next because of the possibility of variations in watershed geometry and the lithology but tends to be a constant throughout the series. The bifurcation ratio is 0.997 which indicates that the geological structures are less disturbing to the drainage pattern (Table 2).

**3.2.4 Stream length ratio ( $R_L$ ) :** The length ratio  $R_L$  is ratio of mean length  $L_u$  of segments of order  $u$  to mean length of segments of the next lower order tends to be constant through the successive orders of watersheds. It indicates that the stream lengths are decreasing with increasing the order of stream. The stream length ratio values for the basin are approximately constant and the mean stream length ratio observed in the area is 0.352 (Table 2).

### 3.3 Aerial aspects of the channel system

**3.3.1 Stream frequency ( $F_s$ ) :** Stream frequency is the total number of streams of all orders per unit area [9]. The stream frequency of Atlur sub-watershed is 2.10, which denotes positive correlation with drainage density value of the area, that is increase in number of streams with respect to increase in drainage density.

**3.3.2 Texture ratio (T) :** Texture ratio is the measure of the closeness of the channel spacing and is related directly to the drainage density. It also depends on underlying lithology, rate of infiltration and relief conditions. In Atlur watershed the texture ratio is 3.80 and is categorized as fine to medium fine in nature.

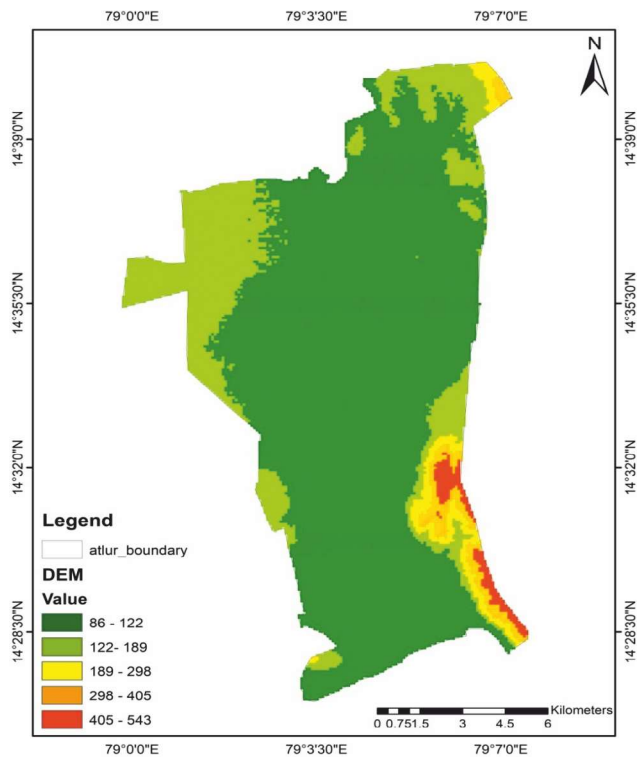
**3.3.3 Basin length ( $L_b$ ) :** There are several definitions defining the length of the basin. For Atlur sub-watershed, the definition of Schumm is taken into consideration [17]. According to him the basin length is the longest dimension of the basin parallel to the principal drainage line. The length of the basin is 23.7 km.

**3.3.4 Elongation ratio ( $R_e$ ) :** Elongation ratio is related to the shape of any drainage basin. Schumm defined elongation ratio as 'the ratio of diameter of a circle of same area as the basin to the maximum basin length' [18]. The elongation ratio in the present study is 0.63 which suggests that the Atlur sub-watershed is less elongated.

**3.3.5 Form factor ( $R_f$ ) :** According to Horton form factor may be defined as the ratio of basin area to square of the basin length [9]. The value of form factor would always be less than 0.754 (for a perfectly circular watershed). Smaller the value of form factor, more elongated will be the watershed. The watershed with high

**Table 2.** Formula for the computation of morphometric parameters

| Morphometric parameters |                                   | Formula/definition                                                                                                                                | Values  | Reference |
|-------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----------|
| Linear                  | Stream order (U)                  | Hierarchical order                                                                                                                                | 378     | [1]       |
|                         | Stream length ( $L_u$ )           | Length of the stream                                                                                                                              | 327.22  | [10]      |
|                         | Mean stream length ( $L_{sm}$ )   | $L_{sm} = L_u / N_u$ ; where $L_u$ = Mean stream length of a given order (km), $N_u$ = Number of stream segments                                  | 0.86    | [10]      |
|                         | Stream length ratio ( $R_L$ )     | $R_L = L_u / L_{u-1}$ where $L_u$ = Total stream length of order L, $L_{u-1}$ = The total stream length of its next lower order                   | 1.00    | [10]      |
|                         | Bifurcation ratio ( $R_b$ )       | $R_b = N_u / N_{u+1}$ where $N_u$ = Number of stream segments present in the given order, $N_{u+1}$ = Number of segments of the next higher order | 0.99    | [17]      |
| Relief                  | Basin relief ( $B_h$ )            | Vertical distance between the lowest and highest points of basin                                                                                  | 23.7 km | [17]      |
|                         | Relief ratio ( $R_h$ )            | $R_h = B_h / L_b$ where $B_h$ = Basin relief, $L_b$ = Basin length                                                                                | 0.30    | [17]      |
|                         | Ruggedness number ( $R_n$ )       | $R_n = B_h \times D_d$ where $B_h$ = Basin relief, $D_d$ = Drainage density                                                                       | 35.07   | [1]       |
| Aerial                  | Drainage density ( $D_d$ )        | $D_d = L/A$ where L = Total length of stream, A = Area of basin                                                                                   | 1.48    | [10]      |
|                         | Stream frequency ( $F_s$ )        | $F_s = N/A$ where N = Total number of stream, A = Area of basin                                                                                   | 2.10    | [10]      |
|                         | Drainage texture ( $D_t$ )        | $D_t = D_d \times F_s$ where $D_d$ = Drainage density, $F_s$ = Stream frequency                                                                   | 3.1     | [12]      |
|                         | Form factor ( $R_f$ )             | $R_f = A/(L_b)^2$ where A = Area of basin, $L_b$ = Basin length                                                                                   | 0.31    | [10]      |
|                         | Circulatory ratio ( $R_c$ )       | $R_c = 4\pi A/P^2$ where A = Area of basin, $\pi = 3.14$ , P = Perimeter of basin                                                                 | 0.36    | [11]      |
|                         | Elongation ratio ( $R_e$ )        | $R_e = \sqrt{(Au/\pi)/L_b}$ where A = Area of basin, $\pi = 3.14$ , $L_b$ = Basin length                                                          | 0.63    | [17]      |
|                         | Length of overland flow ( $L_g$ ) | $L_g = 1/2D_d$ where $D_d$ = Drainage density                                                                                                     | 0.74    | [10]      |



**Figure 3.** DEM map of the study area

form factors has high peak flows of shorter duration, whereas elongated watershed with low form factor ranges from 0.31 indicating them to be elongated in shape and flow for longer duration.

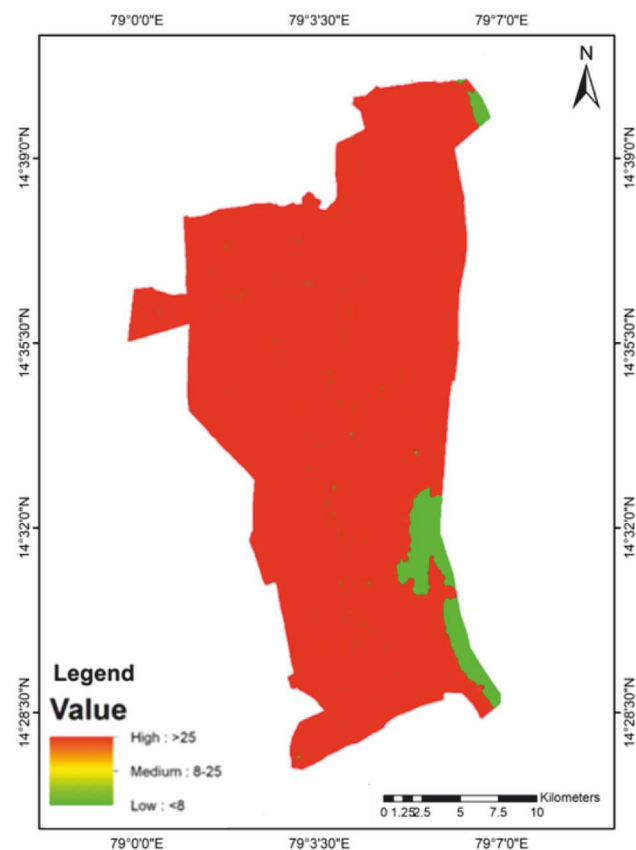
**3.3.6 Circularity ratio ( $R_c$ ):** The outline form of watershed used a dimensionless circularity ratio as a quantitative method [1,11]. Circularity ratio is defined as ratio of watershed area to the area of a circle having same perimeter as the watershed and it is pretentious to lithological character of the watershed. The circularity ratio value (0.36) of the watershed corroborates Miller's range, which indicates that the watershed is elongated in shape, has low discharge of runoff and high permeability of the subsoil condition (Table 2).

**3.3.7 Drainage texture:** Drainage texture is also classified on the basis of density of dissection or texture [15]. Drainage texture is the product of drainage density and stream frequency. It is expressed the same as the drainage and its classification is given in table 3. High relief ratio brings high discharge of surface water in a short duration. Small relief ratio indicates the erosional development of the drainage basin. The drainage area shows coarse grain texture (3.1).



**Table 3. Drainage texture**

| Drainage density | Texture     |
|------------------|-------------|
| < 2              | Very coarse |
| 2 – 4            | Coarse      |
| 4 – 6            | Moderate    |
| 6 – 8            | Fine        |
| > 8              | Very fine   |

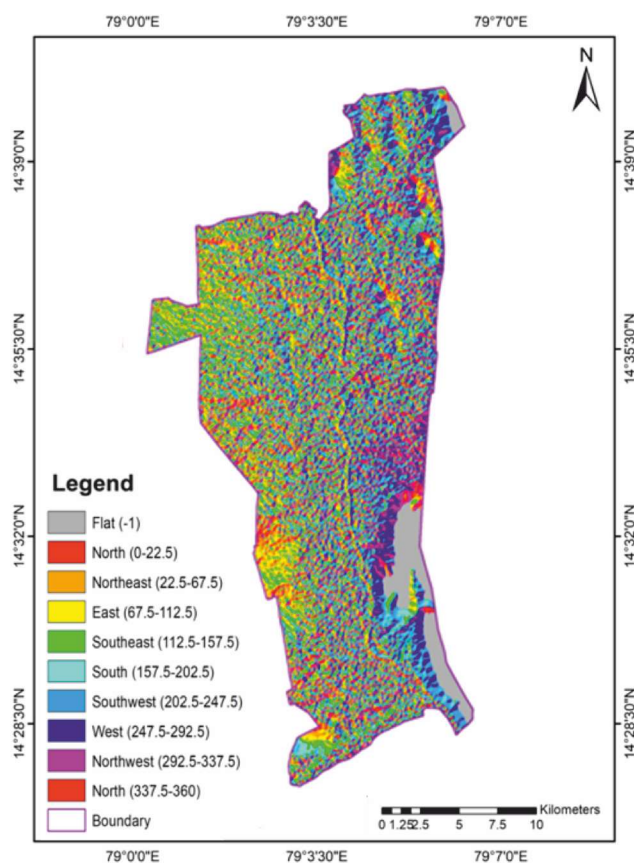


**Figure 4. Slope map of the study area**

**3.3.8 Drainage pattern :** It refers to the orderly spatial arrangement of geologic, topographic or vegetation features. Drainage pattern is an important element in geologic interpretation of aerial photographs. The study area has dendritic to sub-dendritic drainage pattern (Figure 3).

**3.3.9 Drainage frequency :** It is a measure of number of stream segments per unit area and therefore, depends on the stream order, whereas drainage density is independent of stream order.

**3.3.10 Slope:** The slope of terrain to the amount of inclination of physical feature, landform is the horizontal surface. Slope analysis is an important parameter



**Figure 5. Aspect map of the study area**

in morphometric studies. The slope elements, in turn, are controlled by climate morphogenic processes in areas having rocks of varying resistance. According to Burrough, it is the maximum rate of change in value from each cell to its neighbours, is identified as slope grid [20]. The slope exhibited by Atlur sub-watershed varies from 1-25% and slope map is shown in figure 4.

**3.3.11 Aspect:** The aspect of terrain is the direction in which it faces. Aspect influences vegetation type, precipitation patterns, melting of snow and wind exposure. The compass direction of the aspect was derived from the output raster data value. The aspect map of Atlur sub-watershed is shown in figure 5. The east facing slopes mainly occur in the study area, which has higher moisture content and lower evaporation rate and hence has high vegetation index.

**3.3.12 Drainage density ( $D_d$ ):** It is measured as a sum of the channel lengths per unit area and obtained by dividing the total stream length by total area of the basin [1]. Drainage density is controlled by the type of formations in the basin areas with impervious formations will have higher drainage density than those with

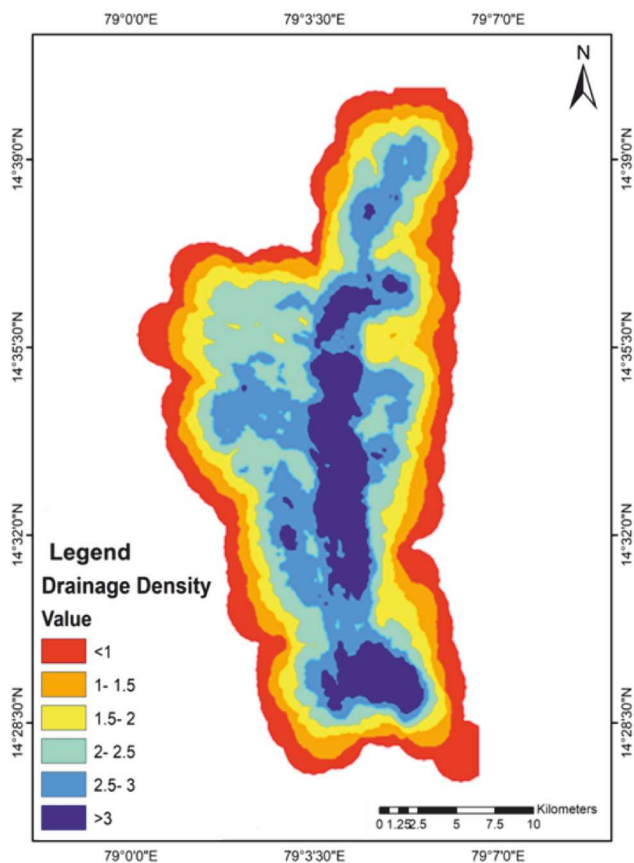


Figure 6. Drainage density map of the study area

pervious formations [16]. In the study area drainage density is low, that is 1.48 which indicates the region is of high resistant or highly permeable sub-soil materials, under dense vegetation cover and where relief is low (Table 2). In general high drainage density is favoured in regions of weak or impermeable surface materials, sparse vegetation and mountainous relief. The drainage density is governed by the factors, like rock type, runoff intensity, soil type, infiltration capacity and percentage of rocky area (Figure 6).

#### 4. CONCLUSION

The present study has proved that the geoprocessing technique used in GIS is an effective tool for computation and analysis of various morphometric parameters of the basin and helps to understand various terrain parameters, such as nature of the bedrock, infiltration capacity, surface runoff, etc. The study area is well drained in nature with the stream order varying from 1 to 5. The area is dominated by lower order streams and the total length of stream segments is maximum in first order streams. Study area is an elongated basin with moderate relief and steep slope due to the

lower elongation ratio (0.63). Stream frequency and drainage density are the prime criteria for the morphometric classification of drainage basins, which certainly control the runoff pattern, sediment yield and other hydrological parameters of the drainage basin. The drainage density appears significantly lower in study area, which is indicative of existence of impermeable rocks and moderate relief. The quantitative analysis of linear, relief and aerial parameters using GIS is found to be of immense utility in area evaluation, area prioritization for soil and water conservation and natural resource management. The geoprocessing techniques employed in this study will assist the planner and decision makers in study area development and management studies.

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